How to use Google Colab

If you want to create a machine learning model but say you don't have a computer that can take the workload, **Google Colab** is the platform for you. In this article, we'll learn how to use google colab.

What is Google Colab?

Google Colab, short for Colaboratory, is a free cloud-based platform provided by Google that allows users to write and execute Python code collaboratively in a Jupyter Notebook environment. **Google Collaboratory notebook,** is designed to facilitate machine learning (ML) and data science tasks by providing a virtual environment, Google colab python with access to free GPU resources.

Benefits of Google Colab

Google Colab offers several benefits that make it a popular choice among data scientists, researchers, and machine learning practitioners. **Key features of Google Collaboratory notebook include:**

- **Free Access to GPUs:** Colab offers free GPU access, which is particularly useful for training machine learning models that require significant computational power.

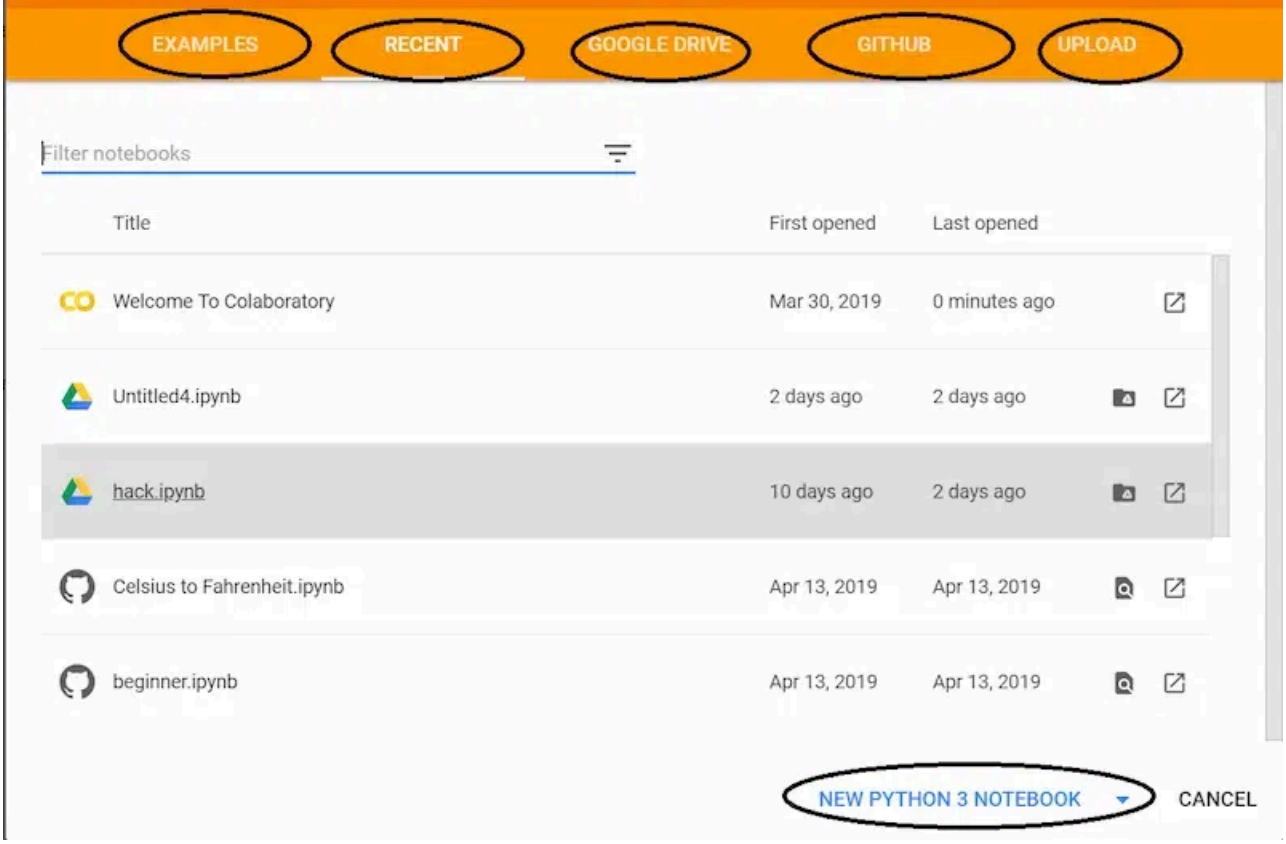
- **No Setup Required** : Colab runs in the cloud, eliminating the need for users to set up and configure their own development environment. This makes it convenient for quick coding and collaboration.
- **Collaborative Editing:** Multiple users can work on the same Colab notebook simultaneously, making it a useful tool for collaborative projects.
- **Integration with Google Drive** : Colab is integrated with Google Drive, allowing users to save their work directly to their Google Drive account. This enables easy sharing and access to notebooks from different devices.
- **Support for Popular Libraries** :Colab comes pre-installed with many popular Python libraries for machine learning, data analysis, and visualization, such as TensorFlow, PyTorch, Matplotlib, and more.
- **Easy Sharing** :Colab notebooks can be easily shared just like Google Docs or Sheets. Users can provide a link to the notebook, and others can view or edit the code in real-time.

Getting Started With Google Colab

To start working with Google Collaboratory Notebook you first need to log in to your Google account, then go to this link https://colab.research.google.com/ .

Open Collaboratory Notebook

On opening the website you will see a pop-up containing the following tabs -



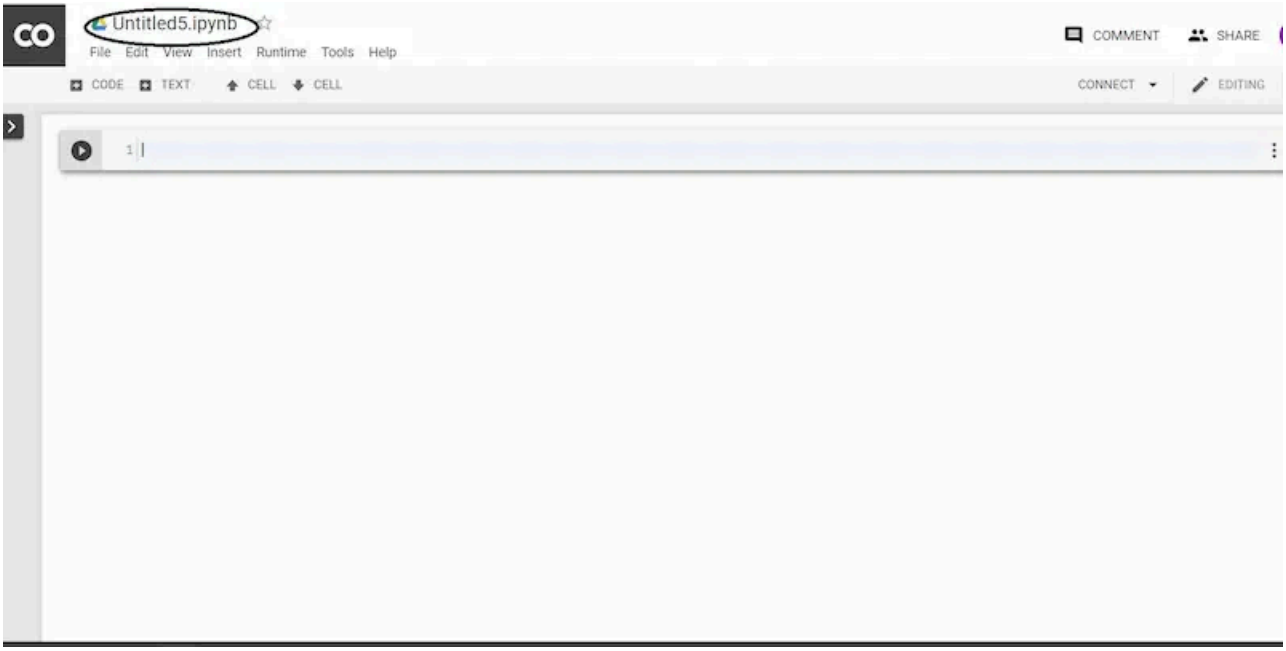
Google Collaboratory Notebook

- **EXAMPLES:** Contain a number of Jupyter notebooks of various examples.
- **RECENT:** Jupyter notebook you have recently worked with.
- **GOOGLE DRIVE:** Jupyter notebook in your google drive.
- **GITHUB:** You can add Jupyter notebook from your GitHub but you first need to connect Colab with GitHub.
- **UPLOAD:** Upload from your local directory.

Create Collaboratory Notebook

Else you can *create a new Jupyter Notebook* by clicking New Python3 Notebook or New Python2 Notebook at the bottom right corner.

Notebook's Description



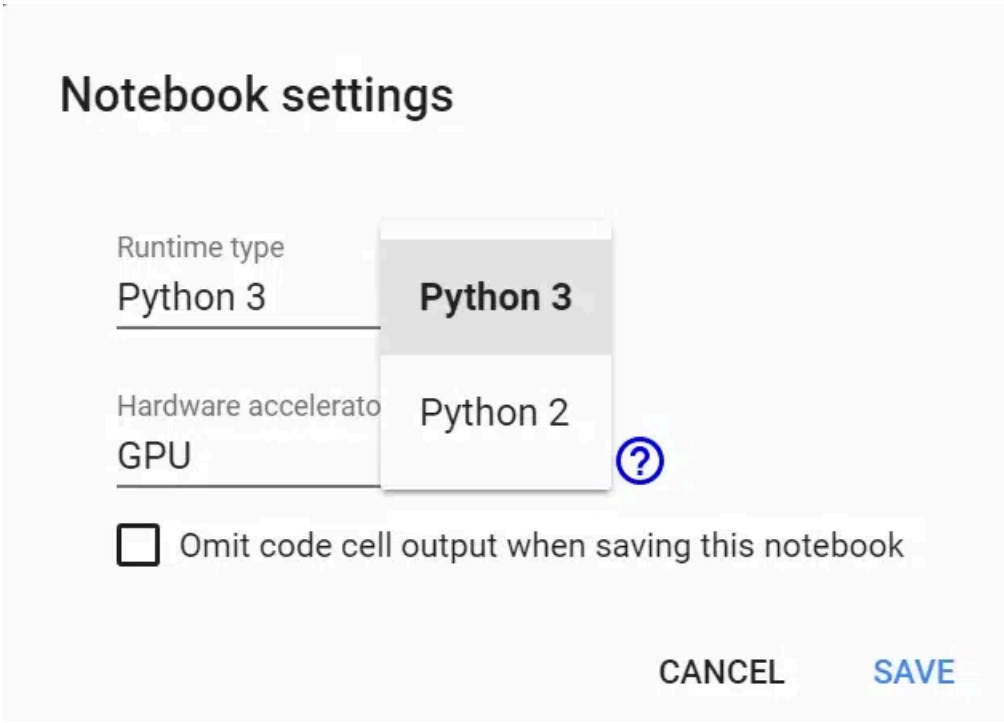
Google Collaboratory Notebook

On creating a new notebook, it will create a Jupyter notebook with Untitled0.ipynb and save it to your google drive in a folder named **Colab Notebooks** .

Now as it is essentially a Jupyter Notebook, all commands of Jupyter Notebooks will work here. Though, you can refer to the details in Getting Started with Jupyter Notebook .

Let's talk about what is different here:

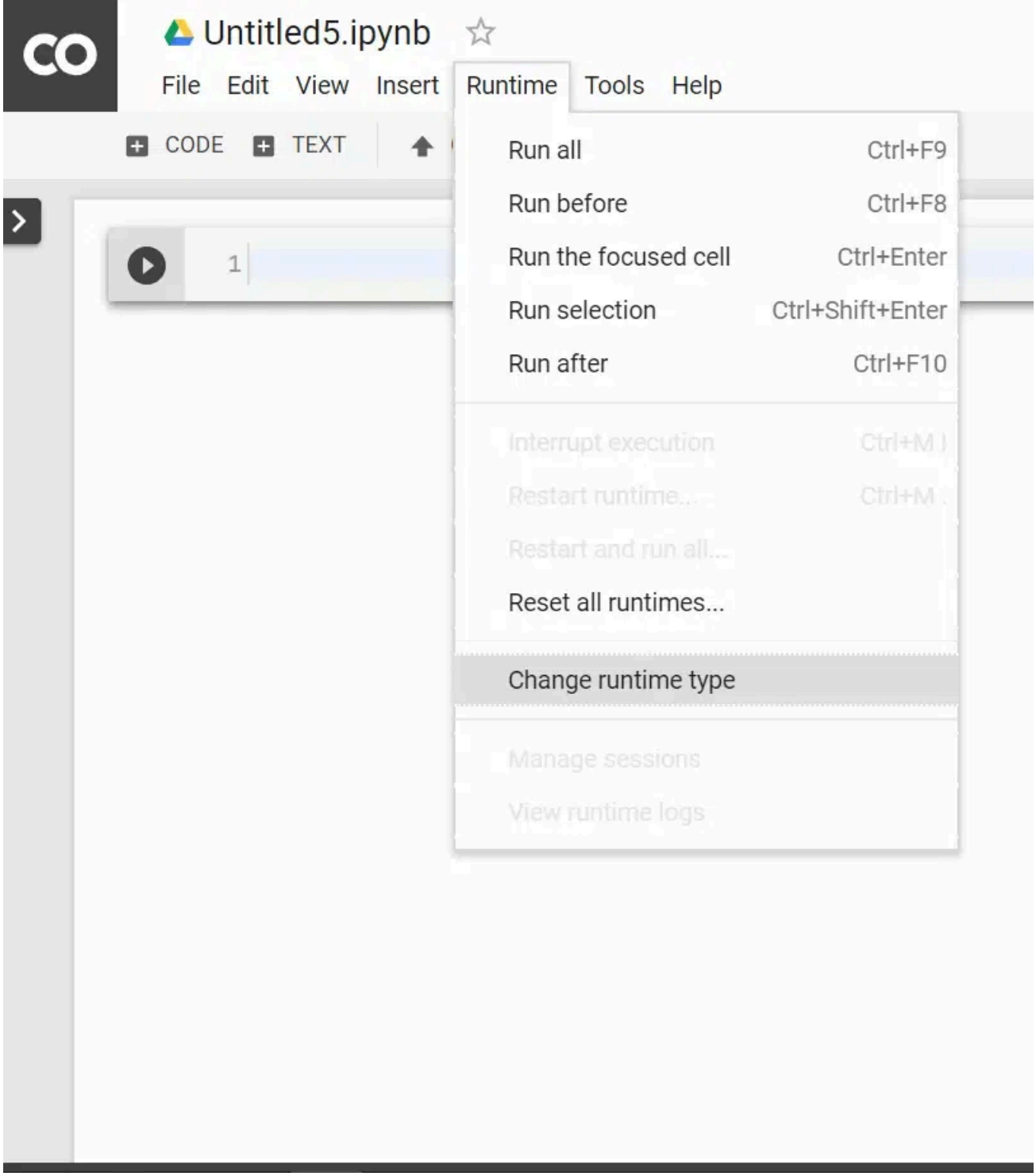
Change Runtime Environment: Click the **"Runtime"** dropdown menu. Select **"Change runtime type"** . Select python2 or 3 from the **"Runtime type"** dropdown menu.



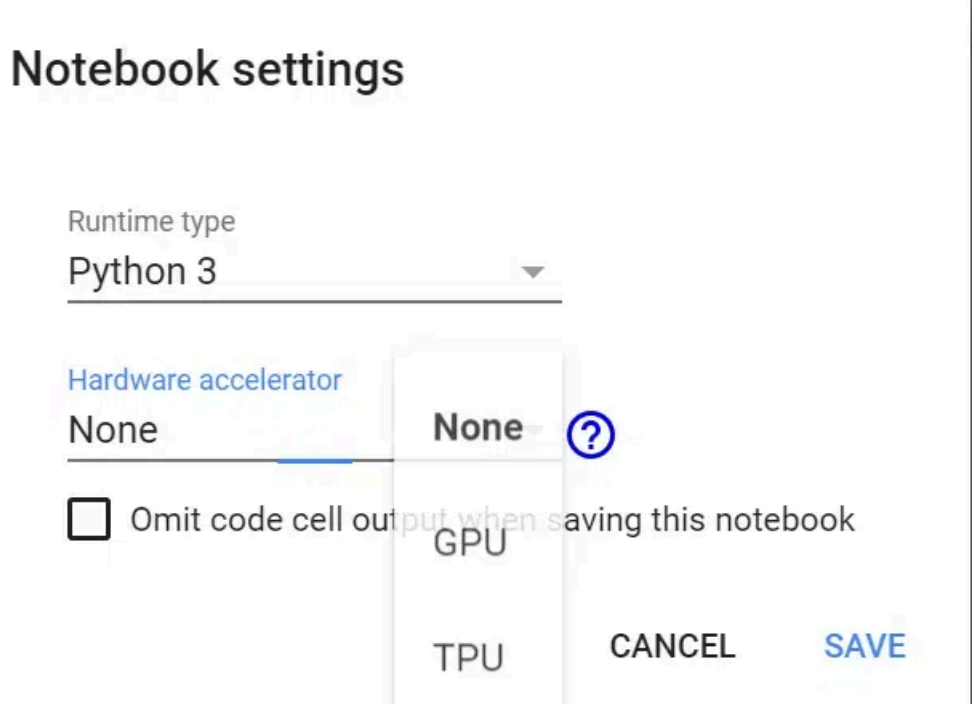
Runtime setting in Google colab

Use GPU and TPU

Click the **"Runtime"** dropdown menu. Select **"Change runtime type"** . Now select anything(GPU, CPU, None) you want in the **"Hardware accelerator"** dropdown menu.



GPU and TPU in Google Colab



Select python in colab

Verify GPU in Colab

```
import tensorflow as tf
tf.test.gpu_device_name()
```

If GPU is connected it will output the following -

```
'/device:GPU:0'
```

Otherwise, it will output following

```
''
```

Verify TPU

```
import os

if 'COLAB_TPU_ADDR' not in os.environ:
  print('Not connected to TPU')
else:
  print(&quot;Connected to TPU&quot;)
```

If GPU is connected it will output following

```
Connected to TPU
```

Otherwise, it will output following

```
Not connected to TPU
```

Install Python packages

Use can use **pip** to install any package. For example:

```
! pip install pandas
```

Clone GitHub repos in Google Colab

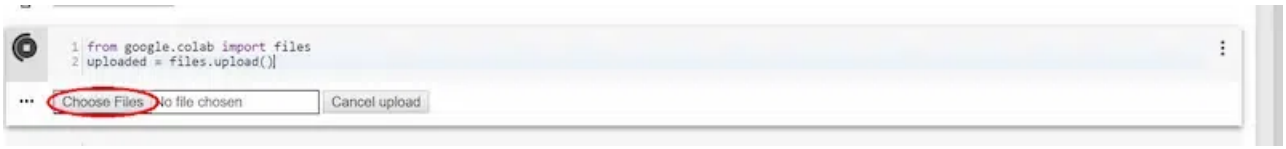
Use the **git clone** command. For example:

```
! git clone https://github.com/souvik3333/Testing-and-Debugging-Tools
```

Upload File on Google Colab

```
from google.colab import files
uploaded = files.upload()
```

Select "Choose file" and upload the file you want. Enable third-party cookies if they are disabled.



Then you can save it in a dataframe.

```
import io
df2 = pd.read_csv(io.BytesIO(uploaded['file_name.csv']))
```

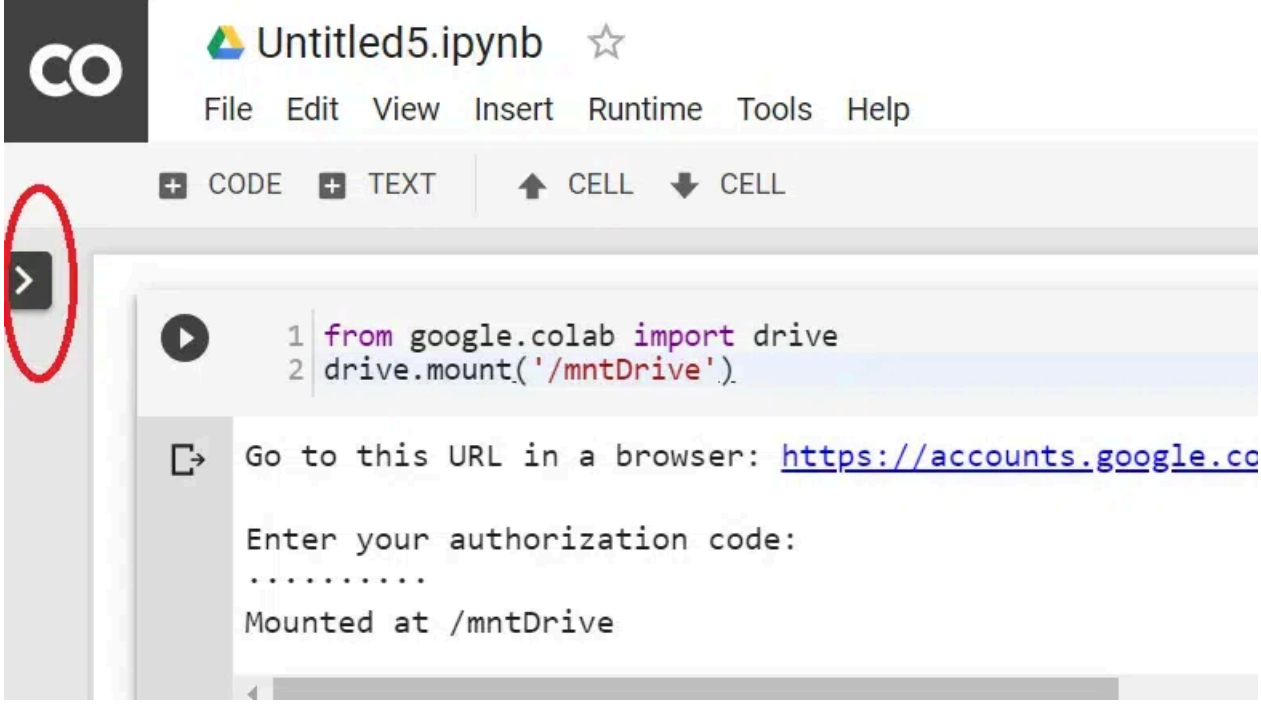
Upload File By Mounting Google Drive

To mount your drive inside the "mntDrive" folder execute the following -

```
from google.colab import drive
drive.mount('/mntDrive')
```

Then you'll see a link, click on the link, then allow access, copy the code that pops up, and paste it at "Enter your authorization code:". Now to see all data in your google drive you need to execute the following:

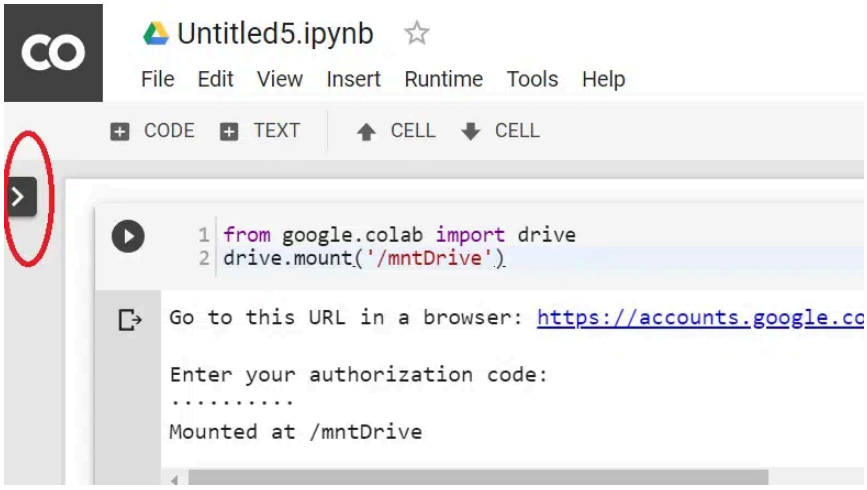
```
! ls '/mntDrive/My Drive&quot'
```

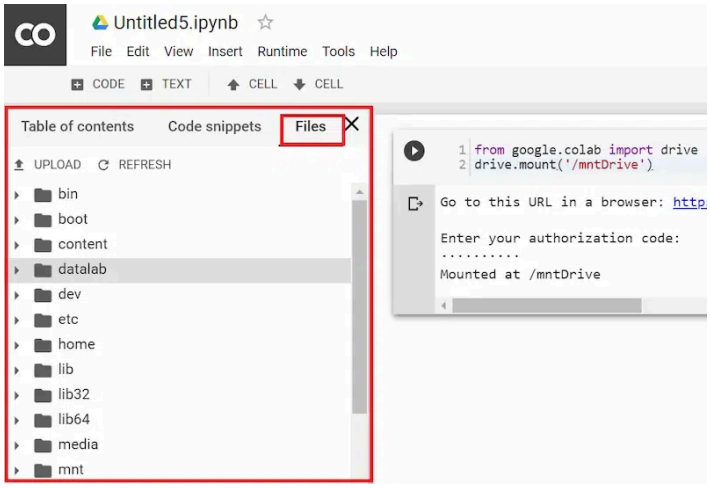


Uploading files on google colab

File Hierarchy In Google Colab

You can also see the file hierarchy by clicking ">" at the top left below the control buttons (CODE, TEXT, CELL).



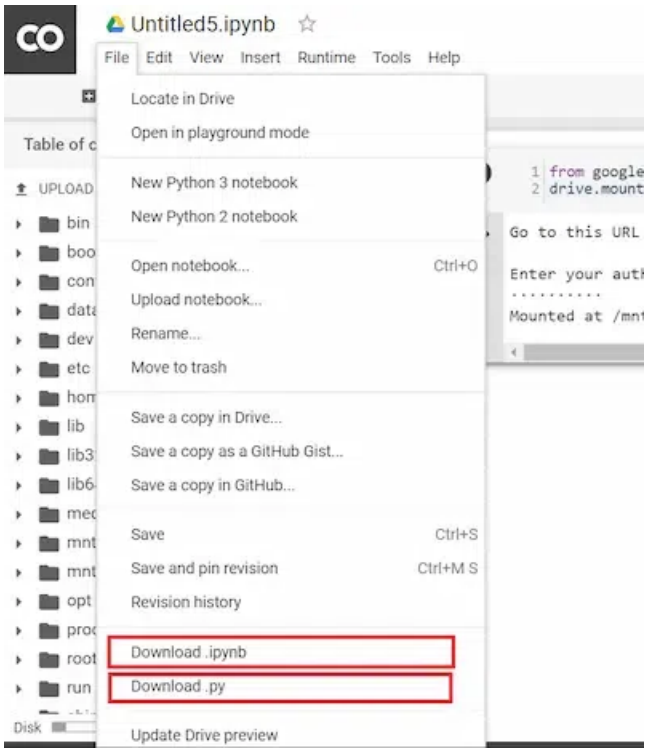


Download Files from Google Colab

Let's say you want to download "file_name.csv". You can copy the file to your google drive (In "data" folder, you need to create the "data" folder in google drive) by executing this:

```
cp file_name.csv &quot;/mntDrive/My Drive/data/renamed_file_name.csv&quot;
```

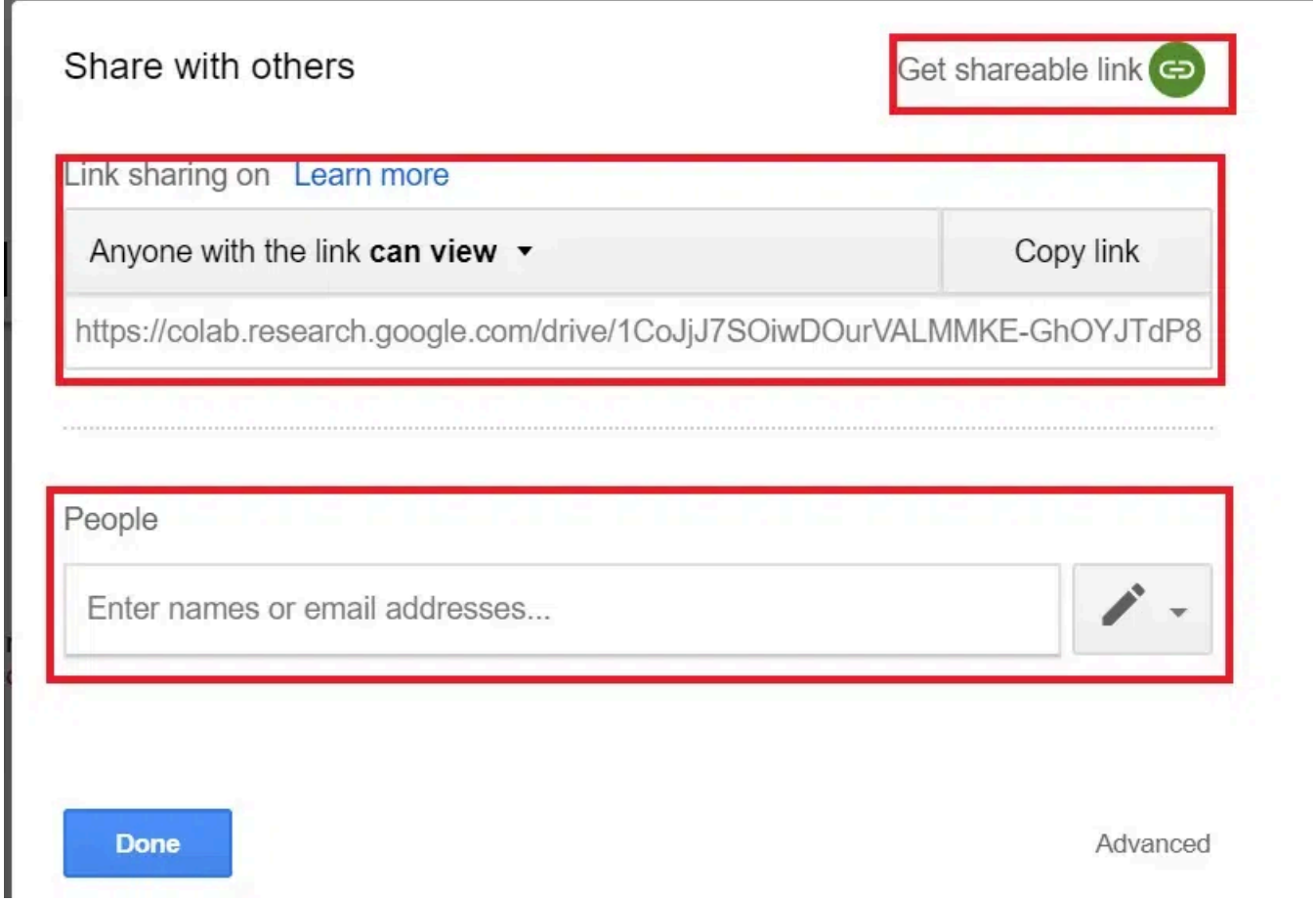
The file will be saved in the "data" folder with the "renamed_file_name.csv" name. Now you can directly download from there, Or, you can just open the file hierarchy and right-clicking will give a download option.
Download Jupyter Notebook: Click the **"File"** dropdown menu at the top left corner. Choose **"download .ipynb"** or **"download .py"**



Downloading files from Google colab

Share Jupyter Notebook: You can share your notebook by adding others' email addresses or by creating a shareable link.

Share jupyter notebook in google colab



Share Google colab Notebook

Conclusion

In conclusion, Google Colab stands out as a versatile and accessible platform for Python coding.

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